

Problems 2

Jakob Sverre Alexandersen
GRA4153 Advanced Statistics

October 9, 2025

Contents

1	Statistics, Sufficiency, Completeness & Ancillarity	2
2	Statistical inference and performance criteria	6
3	Asymptotic statistics	13

1 Statistics, Sufficiency, Completeness & Ancillarity

1. In the context of Example 3.1, which of the following are statistics:

- (i) $\frac{1}{n} \sum_{i=1}^n X_i$
- (ii) $\sum_{i=1}^n X_i$
- (iii) $\frac{1}{n} \sum_{i=1}^n X_i^2$
- (iv) $\frac{1}{n} \sum_{i=1}^n (X_i - \theta)$
- (v) $\frac{1}{n} \sum_{i=1}^n (X_i^2 - 1)$

Example 3.1 states that:

Suppose we observe n samples which satisfy

$$X_i = \theta + \epsilon_i \quad (i = 1, \dots, n), \quad \epsilon_1, \dots, \epsilon_n \text{ are i.i.d. draws from } \mathcal{N}(0, 1), \quad \theta \in \Theta \subset \mathcal{R}$$

Then, our statistical model is

$$\mathcal{P} = \{\mathcal{N}(\theta, 1)^n : \theta \in \Theta\},$$

where $\mathcal{N}(\theta, 1)^n$ denotes the distribution of n i.i.d. draws from a $\mathcal{N}(\theta, 1)$ distribution. $\triangle$

A statistic is any measurable function of the observed sample that does not depend on unknown parameters. Therefore all but (iv) are statistics, as it depends on the parameter θ

2. In the context of Example 3.1, find a minimal sufficient statistic.

A minimal sufficient statistic is $T(X) = \sum_{i=1}^n X_i$

3. Let $X \sim \text{Poisson}(\theta)$ and $Y \sim \text{Poisson}(\theta^2)$, independent, with $\theta \in (0, \infty)$. (i) Find a minimal sufficient statistic for the family of joint distributions for $Z = (X, Y)$. (ii) Is this minimal sufficient statistic complete?

(i) **Minimal sufficient statistic**

The joint pmf is

$$P_\theta(X = x, Y = y) = \frac{e^{-\theta} \theta^x}{x!} \cdot \frac{e^{-\theta^2} \theta^{2y}}{y!} = \frac{1}{x!y!} e^{-(\theta + \theta^2)} \theta^{x+2y}$$

$$P_\theta(x, y) = h(x, y) g_\theta(T), \quad h(x, y) = \frac{1}{x!y!}, \quad g_\theta(t) = e^{-(\theta + \theta^2)} \theta^t, \quad T = X + 2Y$$

By the factorization theorem, $T = X + 2Y$ is sufficient (Fisher-Neyman factorization)

For minimal sufficiency, consider

$$\frac{P_\theta(x, y)}{P_\theta(x', y')} = \frac{x'!y'!}{x!y!} \theta^{(x+2y)-(x'+2y')}$$

The ratio is free of θ iff $x + 2y = x' + 2y'$. Hence $T = X + 2Y$ is minimal sufficient (Lehmann-Scheffé minimal sufficiency criterion)

(ii) Completeness

Let $T = X + 2Y$ for $t = 0, 1, 2, \dots$

$$P_\theta(T = t) = e^{-(\theta + \theta^2)} \theta^t \sum_{y=0}^{\lfloor t/2 \rfloor} \frac{1}{(t - 2y)! y!} = e^{-(\theta + \theta^2)} \theta^t c_t,$$

where $c_t > 0$ (the term $y = 0$ equals $1/t!$)

Assume $\mathbb{E}[g(T)] = 0 \quad \forall \theta > 0$. Then

$$0 = \mathbb{E}[g(T)] = \sum_{t=0}^{\infty} g(t) P_\theta(T = t) = e^{-(\theta + \theta^2)} \sum_{t=0}^{\infty} g(t) c_t \theta^t, \quad \theta > 0$$

Multiplying by $e^{\theta + \theta^2}$ gives

$$\sum_{t=0}^{\infty} a_t \theta^t = 0, \quad a_t = g(t) c_t, \quad \theta > 0$$

By uniqueness of power series coefficients, $a_t = 0 \quad \forall t$, hence $g(t) = 0 \quad \forall t$ since $c_t > 0$. Therefore T is complete (Uniqueness of power series coefficients)

4. Let $Z = (Z_1, \dots, Z_n)$ be i.i.d. standard normal random variables. Show that $\|Z\|$ and $Z/\|Z\|$ are independent. [Hint: consider an appropriate model where each Z_i has a normal distribution depending on some θ and use Basu's Theorem]

Model: for a scale parameter $\sigma > 0$, consider the family $Z \sim \mathcal{N}_n(0, \sigma^2 I_n)$. Define $T = \|Z\|^2$ and $U = Z/\|Z\|$ (defined on $\{Z \neq 0\}$, which has probability 1)

T is complete and sufficient for σ^2 . The joint density factors as $f_\sigma(z) = (2\pi\sigma)^{-n/2} \exp\{-\|z\|^2/(2\sigma^2)\}$, so by the factorization theorem T is sufficient. Moreover, $T \sim \sigma^2 \chi_n^2$, i.e., a Gamma distribution with fixed shape $n/2$ and scale $2\sigma^2$. For this one-parameter Gamma family (fixed shape), the statistic T is complete.

U is ancillary for σ . If $X \sim \mathcal{N}(0, I_n)$, then $Z \stackrel{d}{=} \sigma X$, and $U = Z/\|Z\| = (\sigma X)/\|\sigma X\| = X/\|X\|$, so the distribution of U does not depend on σ (indeed U is uniform on the unit sphere S^{n-1})

By Basu's theorem, a complete sufficient statistic is independent of any ancillary statistic. Hence T and U are independent for every $\sigma > 0$. Since $\|Z\|$ is a measurable function of T , it follows that $\|Z\|$ and $Z/\|Z\|$ are independent.

Taking $\sigma = 1$ gives the desired result for i.i.d. standard normal components. □

5. Consider the family $\{f_p : p \in [0, 1]\}$ where f_p is the Bernoulli pmf of Example 2.2. Show that this is an exponential family and write it in a canonical form.

Example 2.2 states that:

A random variable X has a *Bernoulli(p) distribution* if for some $p \in [0, 1]$

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases},$$

or, equivalently, its pmf is

$$f(x) = \begin{cases} p & x = 1 \\ 1 - p & x = 0 \\ 0 & \text{otherwise} \end{cases}.$$

We write the Bernoulli pmf as

$$f_p(x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}, p \in (0, 1)$$

Then

$$f_p(x) = \exp\{x \log p + (1 - x) \log(1 - p)\} = \exp\left\{x \log \frac{p}{1 - p} + \log(1 - p)\right\}$$

Let the canonical (natural) parameter be

$$\theta = \log \frac{p}{1 - p} \iff p = \frac{e^\theta}{1 + e^\theta},$$

and define the log-partition function

$$A(\theta) = \log(1 + e^\theta)$$

Since $A(\theta) = -\log(1 - p)$, the pmf becomes

$$f_\theta(x) = \exp\{x\theta - A(\theta)\}, \quad x \in \{0, 1\},$$

with base measure $h(x) \equiv 1$ and sufficient statistic $T(x) = x$. The natural parameter space is $\Theta = \mathbb{R}$ (corresponding to $p \in (0, 1)$), with $p = 0, 1$ as limiting cases. Therefore, the Bernoulli family is a one-parameter exponential family in canonical form. $\square$

6. Consider the family $\{p_\lambda : \lambda \in (0, \infty)\}$ where p_λ is the Poisson pmf of Example 2.4. Show that this is an exponential family and write it in canonical form.

Example 2.4 states that:

A random variable X has a Poisson(λ) distribution if it has the pmf

$$f(x) = \frac{\exp(-\lambda)\lambda^x}{x!}, \quad \text{for } x = 0, 1, \dots$$

and x otherwise. Note this places positive mass on every positive integer. The parameter λ is called the *intensity* parameter and the Poisson distribution is often used to model the number of events which occur in a fixed interval.

We write the Poisson pmf as

$$f(x|\lambda) = \frac{\exp(-\lambda)\lambda^x}{x!} = \frac{1}{x!} \exp\{x \log \lambda - \lambda\}, \quad x = 0, 1, 2, \dots, \lambda > 0$$

This has the exponential-family form

$$f(x|\lambda) = h(x) \exp\{\eta(\lambda)T(x) - A(\eta(\lambda))\},$$

with the identifications that the sufficient statistic: $T(x) = x$, canonical parameter: $\eta(\lambda) = \log \lambda \in \mathbb{R}$, log partition: $A(\eta) = \exp(\eta)$ and base measure: $h(x) = 1/x!$

Canonical form(using $\theta = \eta = \log \lambda$):

$$f(x|\theta) = \frac{1}{x!} \exp\{\theta x - e^\theta\}, \quad \theta \in \mathbb{R}, x = 0, 1, \dots$$

Thus, the Poisson family is a (regular) one-parameter exponential family with natural parameter space $\mathbb{R}$

2 Statistical inference and performance criteria

1. Prove the Bias-Variance decomposition in Example 3.10 [Hint: The general case follows from the univariate case which can be shown by applyg $\text{Var}(Z) = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2$ to $Z = T(X) - g(\theta)$]

Example 3.10 [quadratic loss] states that:

Suppose, given some model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, we want to estimate some $g(\theta) \in \mathcal{G} \subset \mathbb{R}^K$. One commonly used loss function is the *quadratic loss*, defined for $t \in \mathbb{R}^K$ as

$$L(\theta, t) = \|g(\theta) - t\|^2.$$

The risk function of an estimate $T = T(X)$ under the quadratic loss is then

$$R(\theta, T) = \mathbb{E}_{P_\theta} \|g(\theta) - T(X)\|^2.$$

This risk function is often called the *mean squared error* (MSE) and satisfies a decomposition into bias and variance terms ("the bias-variance tradeoff"):

$$R(\theta, T) = \mathbb{E}_{P_\theta} \|g(\theta) - T(X)\|^2 = \sum_{k=1}^K \text{Var}_{P_\theta}(T_k(X)) + (\mathbb{E}_{P_\theta} T_k(X) - g_k(\theta))^2$$

Proof:

For any random variable Z and constant a ,

$$\mathbb{E}(Z - a)^2 = \text{Var}(Z) + (\mathbb{E}Z - a)^2$$

Then we write $T = (T_1, \dots, T_K)$ and $g(\theta) = (g_1(\theta), \dots, g_K(\theta))$. Since the squared Euclidean norm is a sum of squares,

$$\mathbb{E}_{P_\theta} \|g(\theta) - T(X)\|^2 = \sum_{k=1}^K \mathbb{E}_{P_\theta} (T_k(X) - g_k(\theta))^2$$

Apply the univariate identity to each component with $Z = T_k(X)$ and $a = g_k(\theta)$:

$$\mathbb{E}_{P_\theta} (T_k(X) - g_k(\theta))^2 = \text{Var}_{P_\theta}(T_k(X)) + (\mathbb{E}_{P_\theta} T_k(X) - g_k(\theta))^2$$

Summing over k gives

$$R(\theta, T) = \sum_{k=1}^K \text{Var}_{P_\theta}(T_k(X)) + (\mathbb{E}_{P_\theta} T_k(X) - g_k(\theta))^2$$

This is the desired bias-variance decomposition (with bias term $\mathbb{E}_{P_\theta} T_k(X) - g_k(\theta)$ for each component). $\square$

2. Let $X = (X_1, \dots, X_n)$ and consider the family $\{p_\lambda^n : \lambda \in (0, \infty)\}$ where p_λ is the Poisson pmf of example 2.4. Consider the estimator $T(X) = X_1$. Show that this is unbiased. Use the Rao-Blackwell Theorem to find a better estimator. [*Derive an explicit expression for the estimator.]

Proof:

Unbiasedness:

Since each $X_i \sim \text{Poisson}(\lambda)$, we have $\mathbb{E}[X_1] = \lambda$. Thus $T(X) = X_1$ is unbiased for λ .

For Rao-Blackwell we let the sufficient statistic for λ be $S = \sum_{i=1}^n X_i$ (by the factorization theorem). Then we define the Rao-Blackwellized estimator $T^*(X) = \mathbb{E}[X_1|S]$. By exchangeability, $\mathbb{E}[X_i|S]$ is the same for all i , and since $\sum_{i=1}^n X_i = S$, taking conditional expectations gives

$$\sum_{i=1}^n \mathbb{E}[X_i|S] = \mathbb{E}\left[\sum_{i=1}^n X_i|S\right] = S,$$

hence $\mathbb{E}[X_1|S] = S/n$. (equivalently, using the standard result that given $S = s$, $(X_1, \dots, X_n)|S = s \sim \text{Multinomial}(s; 1/n, \dots, 1/n)$, so $X_1|S = s \sim \text{Binomial}(s, 1/n)$ and $\mathbb{E}[X_1|S = s] = s/n$)

Therefore, the Rao-Blackwell estimator is

$$\hat{\lambda}_{\text{RB}}(X) = \mathbb{E}[X_1|S] = \frac{S}{n} = \bar{X}$$

with the following properties: □

- Unbiased: $\mathbb{E}[\hat{X}] = \lambda$
- Lower variance: $\text{Var}(X_1) = \lambda$ while $\text{Var}(\hat{X}) = \lambda/n$, so for $n > 1$ the Rao-Blackwell estimator is strictly better
- In fact, since $S \sim \text{Poisson}(n\lambda)$ is complete and sufficient, $\bar{X}$ is the UMVU (uniformly minimum-variance unbiased) estimator of λ

3. In the setting of Example 2.12, show that $a = (n + 2)/(n + 1)$ minimises $R(\theta, \delta_a)$. Calculate the bias of the corresponding estimator δ_a

Example 2.12 [Multivariate Normal] says that:

A random vector $X = (X_1, \dots, X_K)$ has the multivariate normal distribution with mean vector $\mathbb{E}X = \mu$ and a positive definite variance matrix $\mathbb{E}(X - \mu)(X - \mu)' = \Sigma$ if its joint pdf is

$$f(x) = \det(2\pi\Sigma)^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)'\Sigma^{-1}(x - \mu)\right)$$

If $X = (X_1, \dots, X_k)$ is multivariate normally distributed, then each X_k is marginally normally distributed. The converse is not necessarily true. It does hold in the special case that $X_1, \dots, X_K$ are independent.

Q: What is the estimator here? Which risk function are we using?

4. In the setting of Example 2.13 show that conditional on $T = t, X_1 = t$ with probability $1/n$ and X_1 is uniformly distributed on $(0, t)$ with probability $(n - 1)/n$

Example 2.13 says that:

Let X_n be a random variable with $P(X_n = n) = 1/n$ and $P(X_n = 0) = 1 - 1/n$. Then $X_n \xrightarrow{P} 0$ (hence also weakly) [Exercise] but $\mathbb{E}X_n = n \times 1/n = 1$ for each n .

Let $X_1, \dots, X_n$ be i.i.d. Uniform(0, 1) and let $T = \max\{X_1, \dots, X_n\}$. Dix $0 < t < 1$. We compute the regular conditional law of X_1 given $T = t$ by a small-interval argument.

Denominator: For small $h > 0$:

$$P(t < T \leq t + h) = P(\text{at least one } X_i \in (t, t + h], \text{ others } \leq t) = nt^{n-1}h + o(h),$$

since the density of T at t is nt^{n-1} .

Continuous part on $(0, t)$: For $0 \leq x < t$,

$$P(X_1 \leq x, t < T \leq t + h) = P(X_1 \leq x, \max(X_2, \dots, X_n) \in (t, t + h]) = (n - 1)xt^{n-2}h + o(h),$$

because $X_1 \leq x$ contributes factor x , one of $X_2, \dots, X_n$ lands in $(t, t + h]$ (probability $\approx (n - 1)t^{n-2}h$), and the change of two or more in $(t, t + h]$ is $o(h)$

Hence, for $0 \leq x < t$,

$$P(X_1 \leq x | T \in (t, t + h]) \rightarrow \frac{(n - 1)xt^{n-2}}{nt^{n-1}} = \frac{n - 1}{n} \frac{x}{t} \text{ as } h \rightarrow 0$$

Thus the conditional law of X_1 on $(0, t)$ has density $\frac{n-1}{n} \cdot \frac{1}{t}$, i.e., Uniform(0, t) with weight $(n - 1)/n$

Atom at t : Also,

$$P(X_1 \in (t, t + h], t < T \leq t + h) = P(X_1 \in (t, t + h], X_2, \dots, X_n \leq t) = t^{n-1}h + o(h),$$

so

$$P(X_1 \in (t, t + h] | T \in (t, t + h]) \rightarrow \frac{t^{n-1}}{nt^{n-1}} = \frac{1}{n}.$$

This yields $P(X_1 = t | T = t) = 1/n$

Therefore, conditional on $T = t, X_1$ equals t with probability $1/n$, and with the remaining probability $(n - 1)/n$ it is Uniform(0, t). $\square$

5. Consider a random sample $X = (X_1, \dots, X_n)$ where each $X_i \sim \mathcal{N}(\mu, \sigma^2)$. Calculate the information matrix $I(\theta)$ where $\theta = (\mu, \sigma^2)$ and obtain UMVU estimators for μ and σ^2 respectively.

We need to find the log-likelihood for one observation:

$$\ell(\mu, \sigma^2; x) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x - \mu)^2}{2\sigma^2}$$

For the sample of size n :

$$\ell(\mu, \sigma^2; X) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

Scores:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu), \quad \frac{\partial \ell}{\partial (\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (X_i - \mu)^2$$

Expected Fisher information matrix $I(\theta) = \mathbb{E}[\text{Hessian}(\ell)]$:

$$\begin{aligned} I_{\mu\mu} &= -\mathbb{E} \left[\frac{\partial^2 \ell}{\partial \mu^2} \right] = \frac{n}{\sigma^2} \\ I_{\mu, \sigma^2} &= -\mathbb{E} \left[\frac{\partial^2 \ell}{\partial \mu \partial (\sigma^2)} \right] = 0 \\ I_{\sigma^2, \sigma^2} &= -\mathbb{E} \left[\frac{\partial^2 \ell}{\partial (\sigma^2)^2} \right] = \frac{n}{2\sigma^4} \end{aligned}$$

Thus,

$$I(\theta) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix}$$

UMVU estimators:

- The statistic $(\bar{X}, S^2)$ with $\bar{X} = \frac{1}{n} \sum X_i$ and $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ is sufficient and complete for (μ, σ^2) (for $n \geq 2$) in the normal family.
- Since $\mathbb{E}[\bar{X}] = \mu$ and $\mathbb{E}[S^2] = \sigma^2$, by Lehmann-Scheffé, the UMVU estimators are
 - for $\mu : \hat{\mu}_{\text{UMVU}} = \bar{X}$
 - for $\sigma^2 : \widehat{\sigma^2}_{\text{UMVU}} = S^2$

□

6. Suppose p_θ is a pdf for each θ , that (i) of Theorem 3.7 holds and that p_θ can be twice differentiated under the integral sign in that

$$\nabla_\theta \nabla_{\theta'} \int p_\theta(x) dx = \int \nabla_\theta \nabla_{\theta'} p_\theta(x) dx$$

Show that $-I(\theta) = \mathbb{E}_{P_\theta}[\nabla_{\theta'} \ell_\theta]$. This is called the *information equality*.

Theorem 3.7 [The information inequality] states:

Suppose that X has a distribution in the dominated model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ with $\Theta \in \mathbb{R}^K$ let $\ell_\theta := \nabla_\theta \log p_\theta$. If δ is any (real-valued) statistic with $\mathbb{E}_{P_\theta} \delta(X)^2 < \infty$ and:

(i) ℓ_θ exists, $\mathbb{E}_{P_\theta} \ell_\theta(X) = 0$ and $I(\theta) := \mathbb{E}_{P_\theta}[\ell_\theta(X) \ell_\theta(X)']$ exists and is positive definite,

(ii) $\nabla_\theta \mathbb{E}_{P_\theta}[\delta(X)]$ exists and satisfies $\nabla_\theta \mathbb{E}_{P_\theta}[\delta(X)] = \int \nabla_\theta \delta(x) p_\theta(x) dx$,

then

$$\text{Var}_{P_\theta}(\delta(X)) \geq (\nabla_\theta \mathbb{E}_{P_\theta}[\delta(X)])' I(\theta)^{-1} (\nabla_\theta \mathbb{E}_{P_\theta}[\delta(X)])$$

Proof:

Let $\ell_\theta(x) := \nabla_\theta \log p_\theta(x)$. For each x with $p_\theta(x) > 0$, the Hessian of the log-density satisfies the standard identity (by the quotient rule for vector derivatives):

$$\nabla_\theta \ell_\theta(x) = \nabla_{\theta'} \nabla_\theta \log p_\theta(x) = \frac{\nabla_{\theta'} \nabla_\theta p_\theta(x)}{p_\theta(x)} = \frac{\nabla_\theta p_\theta(x) \nabla_{\theta'} p_\theta(x)}{p_\theta(x)^2} = \frac{\nabla_{\theta'} \nabla_\theta p_\theta(x)}{p_\theta(x)} = -\ell_\theta(x) \ell_\theta(x)'$$

Taking expectation under P_θ and using the given interchange of differentiation and integration,

$$\mathbb{E}_{P_\theta}[\nabla_{\theta'} \ell_\theta(X)] = \int \left(\frac{\nabla_{\theta'} \nabla_\theta p_\theta(x)}{p_\theta(x)} - \ell_\theta(x) \ell_\theta(x)' \right) p_\theta(x) dx = \int \nabla_{\theta'} \nabla_\theta p_\theta(x) dx - \mathbb{E}[\ell_\theta(X) \ell_\theta(X)']$$

By normalization and the assumed twice-differentiation under the integral sign,

$$\int \nabla_{\theta'} \nabla_\theta p_\theta(x) dx = \nabla_{\theta'} \nabla_\theta \int p_\theta(x) dx = \nabla_{\theta'} \nabla_\theta 1 = 0$$

Hence

$$\mathbb{E}_{P_\theta}[\nabla_{\theta'} \ell_\theta(X)] = -\mathbb{E}[\ell_\theta(X) \ell_\theta(X)'] = -I(\theta),$$

which is the information equality □

7. Suppose that $S(X)$ is a confidence set for $g(\theta)$ s.t. $P_\theta(g(\theta) \in S(X)) = 1 - \alpha \quad \forall \theta \in \Theta$ Form a test, φ , as in (21). What is the size of φ ?

The test in (21) is described as:

$$\varphi(x) := \begin{cases} 1 & \text{when } g_0 \notin S(X) \\ 0 & \text{otherwise} \end{cases}$$

Size of $\varphi : \alpha$

Proof:

For any θ with $g(\theta) = g_0$,

$$P_\theta(\varphi(X) = 1) = P_\theta(g_0 \notin S(X)) = 1 - P_\theta(g_0 \in S(X)) = 1 - P_\theta(g(\theta) \in S(X)) = \alpha,$$

since $S(X)$ has exact coverage $1 - \alpha \quad \forall \theta$. Thus

$$\sup_{\theta: g(\theta) = g_0} P_\theta(\varphi(X) = 1) = \alpha$$

□

8. Show: k_α is the $1 - \alpha$ quantile of the $\chi^2(1)$ distribution iff $k_\alpha^{1/2}$ is the $1 - \alpha/2$ quantile of the $\mathcal{N}(0, 1)$ distribution.

Proof:

Let $Z \sim \mathcal{N}(0, 1)$ and note that $Z^2 \sim \chi^2(1)$

For $k \geq 0$,

$$P(\chi^2(1) \leq k) = P(Z^2 \leq k) = P(-\sqrt{k} \leq Z \leq \sqrt{k}) = \Phi(\sqrt{k}) - \Phi(-\sqrt{k}) = 2\Phi(\sqrt{k}) - 1,$$

where Φ is the standard normal cdf.

Thus, k_α is the $1 - \alpha$ quantile of $\chi^2(1)$ iff

$$2\Phi(\sqrt{k_\alpha}) - 1 = 1 - \alpha \iff \Phi(\sqrt{k_\alpha}) = 1 - \alpha/2$$

Since Φ is strictly increasing, this is equivalent to

$$\sqrt{k_\alpha} = \Phi^{-1}(1 - \alpha/2),$$

i.e., $\sqrt{k_\alpha}$ is the $1 - \alpha/2$ quantile of $\mathcal{N}(0, 1)$

□

9. Verify the form of $S(X)$ in Example 3.15

Example 3.15 states:

Suppose we are interested in the mean of $X \sim \mathcal{N}(\mu, \sigma^2)$ with known σ^2 . In particular, let's suppose we want to construct a $1 - \alpha$ confidence set for μ . The set $S(X) = \{a : \phi_a(X) = 0\}$ is a $1 - \alpha$ confidence set for μ . Let's describe this set in a more straightforward way. Note that

$$\phi_a(X) = 0 \iff T \leq k_\alpha \iff T^{1/2} \leq k_\alpha^{1/2} \text{ or } -T^{1/2} \leq k_\alpha^{1/2}$$

Here we note that $c_\alpha := k_\alpha^{1/2}$ is the $1 - \alpha/2$ quantile of a standard normal r.v. Since $T^{1/2} = \frac{(X-a)}{\sigma}$ this gives our set as

$$S(X) = [X - \sigma c_\alpha, X + \sigma c_\alpha]$$

Claim: for $X \sim \mathcal{N}(\mu, \sigma^2)$ with known σ^2 , the inverted acceptance region yields

$$S(X) = \{a : \phi_a(X) = 0\} = [X - \sigma c_\alpha, X + \sigma c_\alpha]$$

where $c_\alpha := k_\alpha^{1/2}$ is the $1 - \alpha/2$ standard normal quantile

Proof:

Under $H_0 : \mu = a$, the standardized statistic

$$Z = \frac{X - a}{\sigma} \sim \mathcal{N}(0, 1)$$

so $T = Z^2 = \frac{(X-a)^2}{\sigma^2} \sim \chi_1^2$ [standard result: if $Z \sim \mathcal{N}(0, 1)$ then $Z^2 \sim \chi_1^2$]

A level- α two-sided test based on T rejects T for $T > k_\alpha$, where k_α is the $(1 - \alpha)$ quantile of χ_1^2 . Hence

$$\phi_a(X) = 0 \iff T \leq k_\alpha \iff \frac{(X-a)^2}{\sigma^2} \leq k_\alpha \iff |X-a| \leq \sigma \sqrt{k_\alpha}$$

Let $c_\alpha = \sqrt{k_\alpha}$. Since $Z \sim \mathcal{N}(0, 1)$,

$$\mathbb{P}(|Z| \leq c_\alpha) = \mathbb{P}(Z^2 \leq k_\alpha) = 1 - \alpha \iff 2\Phi(c_\alpha) - 1 = 1 - \alpha \iff \Phi(c_\alpha) = 1 - \frac{\alpha}{2}$$

so $c_\alpha = z_{1-\alpha/2}$. Therefore

$$\phi_a(X) = 0 \iff -\sigma c_\alpha \leq X - a \leq \sigma c_\alpha \iff a \in [X - \sigma c_\alpha, X + \sigma c_\alpha]$$

which verifies the standard form. □

3 Asymptotic statistics

1. Prove Proposition 3.4 [Hint: Example 3.10 may be useful]

Proposition 3.4 states:

If $(\hat{\delta}_n)_{n \in \mathbb{N}}$ is a sequence of estimators s.t. $b(\theta) := \mathbb{E}_{P_\theta} \hat{\delta}_n - g(\theta) \rightarrow 0$ and $\text{Var}_{P_\theta}(\hat{\delta}_n) \rightarrow 0$ as $n \rightarrow \infty$ for each $\theta \in \Theta$, then $\hat{\delta}_n \xrightarrow{P_\theta} g(\theta)$

Example 3.10 [quadratic loss] states that:

Suppose, given some model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, we want to estimate some $g(\theta) \in \mathcal{G} \subset \mathbb{R}^K$. One commonly used loss function is the *quadratic loss*, defined for $t \in \mathbb{R}^K$ as

$$L(\theta, t) = \|g(\theta) - t\|^2.$$

The risk function of an estimate $T = T(X)$ under the quadratic loss is then

$$R(\theta, T) = \mathbb{E}_{P_\theta} \|g(\theta) - T(X)\|^2.$$

This risk function is often called the *mean squared error* (MSE) and satisfies a decomposition into bias and variance terms ("the bias-variance tradeoff"):

$$R(\theta, T) = \mathbb{E}_{P_\theta} \|g(\theta) - T(X)\|^2 = \sum_{k=1}^K \text{Var}_{P_\theta}(T_k(X)) + (\mathbb{E}_{P_\theta} T_k(X) - g_k(\theta))^2$$

Proof: (using the bias-variance decomposition and Markov/Chebyshev)

Fix any $\theta \in \Theta$ and write $b_n(\theta) := \mathbb{E}_{P_\theta} \hat{\delta}_n - g(\theta)$. By the quadratic-loss risk decomposition, for $K = 1$,

$$\mathbb{E}_{P_\theta} (\hat{\delta}_n - g(\theta))^2 = \text{Var}_{P_\theta}(\hat{\delta}_n) + (b_n(\theta))^2$$

By assumption, $\text{Var}_{P_\theta}(\hat{\delta}_n) \rightarrow 0$ and $b_n(\theta) \rightarrow 0$, hence

$$\mathbb{E}_{P_\theta} (\hat{\delta}_n - g(\theta))^2 \longrightarrow 0$$

For any $\epsilon > 0$, apply Markov's inequality to the nonnegative r.v. $(\hat{\delta}_n - g(\theta))^2$:

$$P_\theta(|\hat{\delta}_n - g(\theta)| > \epsilon) \leq \frac{\mathbb{E}(\hat{\delta}_n - g(\theta))^2}{\epsilon^2} \longrightarrow 0$$

which is exactly $\hat{\delta}_n \xrightarrow{P_\theta} g(\theta)$ (Markov's inequality) □

2. Suppose that $X_1, \dots, X_n$ are i.i.d. $\text{Poisson}(\lambda)$ random variables. Show that $\hat{\lambda}_n = \frac{1}{n} \sum_{i=1}^n X_i$ is consistent for λ .

Proof:

Let $\hat{\lambda}_n = \frac{1}{n} \sum_{i=1}^n X_i$ with $X_i \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$. Then

$$\begin{aligned}\mathbb{E}[\hat{\lambda}_n] &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \lambda \\ \text{Var}(\hat{\lambda}_n) &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n\lambda = \frac{\lambda}{n}\end{aligned}$$

For any $\epsilon > 0$, by Chebyshev's inequality,

$$P(|\hat{\lambda}_n - \lambda| \geq \epsilon) \leq \frac{\text{Var}(\hat{\lambda}_n)}{\epsilon^2} = \frac{\lambda}{n\epsilon^2} \rightarrow 0$$

Hence, $\hat{\lambda}_n \xrightarrow{P} \lambda$, so $\hat{\lambda}_n$ is consistent for λ (Chebyshev inequality). $\square$

3. Draw some large integer n samples from (i) a standard normal distribution and (ii) from a Cauchy distribution. Plot the cumulative sample average over $1, \dots, n$ (i.e. plot $X_1, (X_1 + X_2)/2, (X_1 + X_2 + X_3)/3, \dots, n^{-1} \sum_{i=1}^n X_i$). Describe the observed behaviour and whether the sequence appears to follow the WLLN. If not, why not?

Normal case: The running mean $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$ stabilizes near 0 and the fluctuation shrinks and grows.

This is consistent with the Weak Law of Large Numbers: For i.i.d. variables with finite means, $\bar{X} \xrightarrow{P} \mathbb{E}[X]$. For $X \sim \mathcal{N}(0, 1)$, $\mathbb{E}[X] = 0$, so the WLLN holds. In fact, by the Strong Law, the convergence is a.s., and typical deviations are on the order of $1/\sqrt{n}$ by the CLT.

Cauchy case: The running mean does not settle down; instead it exhibits occasional large jumps that can dramatically change the average even at large n . The WLLN requires a finite expectation, but the Cauchy distribution has no mean (the expectation is undefined), so the WLLN does not apply. In fact, the Cauchy is 1-stable, so for i.i.d. standard Cauchy variables, the average $\bar{X}_n$ has the same Cauchy distribution for every n , and therefore does not concentrate around any constant and does not converge in probability.

